Mini-Project 2: Interactive Dashboard

Proposal + Sketches

Where, When, and How High Do Wildlife Strikes Hurt Aircraft?

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Assignment: Mini-Project 2 - Proposal + Sketches

Collaborators: None

AI disclosure. Tool: Claude Opus 5.5 Web
How it was used: Proposal grammar was refined to be more concise.

1 Dataset

FAA Wildlife Strike Database: The database holds about 344,000 U.S. strikes from 1990–2025 across 103 fields. Each row is one reported strike between wildlife and a civil aircraft.

Why this dataset: Wildlife striking aircraft, particularly civilian is a real and growing safety problem, thus $400M of airport funding has gone into wildlife mitigation. The data also brings up questions that summary statistics cannot answer. Most strikes are harmless, so the interesting structure is in *which* strikes cause damage, at what height, in which phase of flight, by which animals, and at which airports.

Scope: U.S. airports, 2000-2025: 307,820 strikes, using the following fields:

Field (FAA name)	Type	Values / notes
`INCIDENT_DATE`	Quantitative	Aggregated to year–month; 312 months.
`HEIGHT`	Quantitative	Feet above ground level; binned to 0, 1–500, 501–1,000, then 1,000-ft bands up to 5,000, and 5,001+ (8 bins).
`DAMAGE_LEVEL`	Ordinal	N (none) $<$ M (minor) $<$ S (substantial) $<$ D (destroyed); M? = damaged but extent is unknown.
`PHASE_OF_FLIGHT`	Ordinal	Taxi $\rightarrow$ Take-off run $\rightarrow$ Climb $\rightarrow$ En route $\rightarrow$ Descent $\rightarrow$ Approach $\rightarrow$ Landing roll. Vague entries (Arrival, Departure, Local, Parked which account for 2%) count as unknown.
`SPECIES`	Categorical	945 species, grouped into 8 groups (waterfowl, gulls/terns, raptors, doves/pigeons, perching birds, shorebirds, bats, terrestrial mammals), "Unidentified" and "Other".
`SIZE`	Ordinal	Small $<$ Medium $<$ Large.
`TIME_OF_DAY`	Ordinal	Dawn, Day, Dusk, Night.
`AIRPORT_ID`	Categorical + spatial	ICAO code. Airport coordinates are joined from the public OurAirports list by ICAO code and aggregated per airport.
`COST_REPAIRS_INFL_ADJ`	Quantitative	Inflation-adjusted USD; only 1.5% of strikes report it, so it is shown as "reported repair cost" with the number of reports.

Reporting is voluntary, so rising counts partly reflect better reporting. Many fields are incompleteand unknown values are kept and shown as their own "Unknown" category, and damage rates are computed only over strikes whose damage is known. The map omits strikes with no recorded airport and the dashboard excludes strikes at foreign airports.

2 Research Questions

RQ1: How high, and in which phase? How are strikes spread across height above ground and phase of flight, and are strikes above 500 ft more likely to cause damage than those near the ground?
(We look at height profile (V2) and phase bars (V3), each stacked by damage level.)

RQ2: What hits? Which wildlife groups cause more damage than their share of strikes would suggest (e.g., few strikes but a high damage rate), and does that change between day and night?
(Species view (V5) with the time-of-day and animal-size filters sheds insight.)

RQ3: When and where? How has monthly strike volume and the damaging share changed from 2000 to 2025, how strong is the late-summer peak, and which airports stand out for volume versus damage rate?
(Answered by timeline (V4) and airport map (V1).)

3 Layout Sketches

I sketched three layouts as wireframes in Figma and chose **Option A**. Each layout is shown below: Option A with numbered annotations (Fig. 1), Option B (Fig. 2), and Option C (Fig. 3). The wireframes are greyscale: boxes for regions, an X-box for the basemap, grey bars for unlabeled text, and outlined placeholder marks. Each mark shows the chart *type* only, not data. Color is described in the annotations rather than drawn, so the layout currently portrays the structure alone.

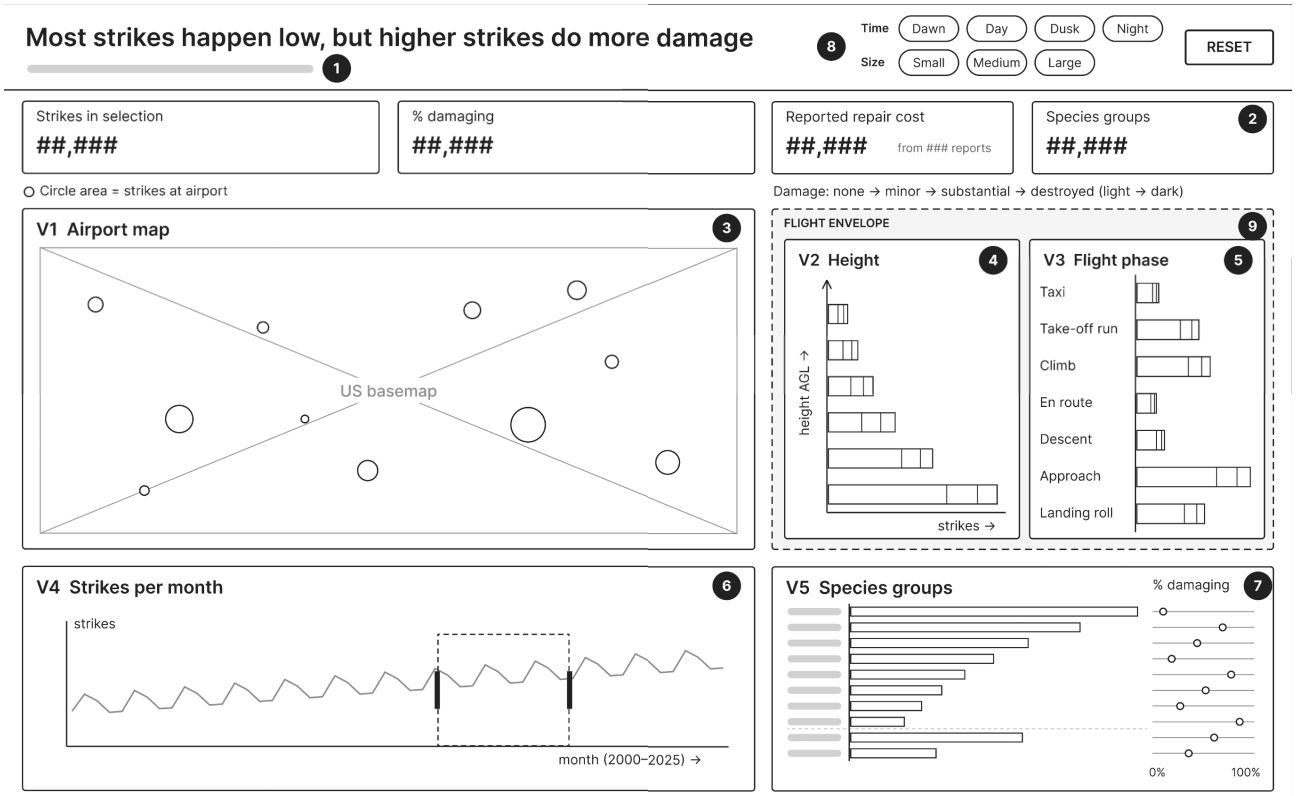


Figure 1: Option A (chosen): overview-first grid, annotated. Numbered callouts are explained below.

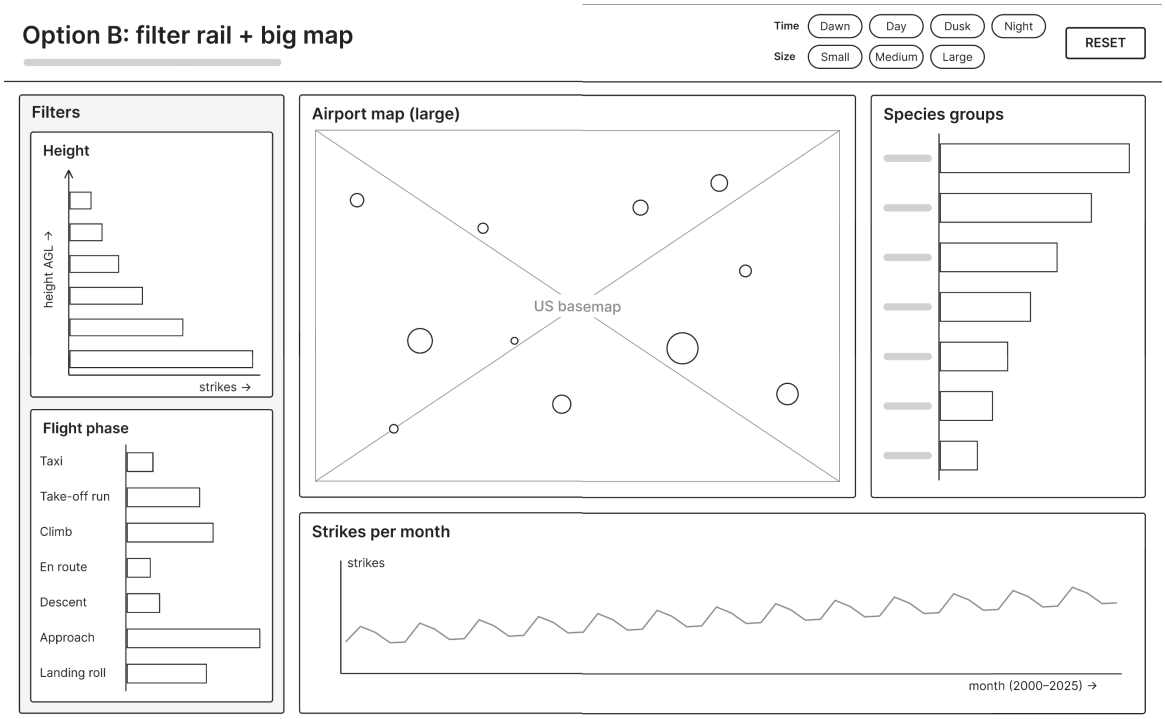


Figure 2: Option B: filter rail + large map.

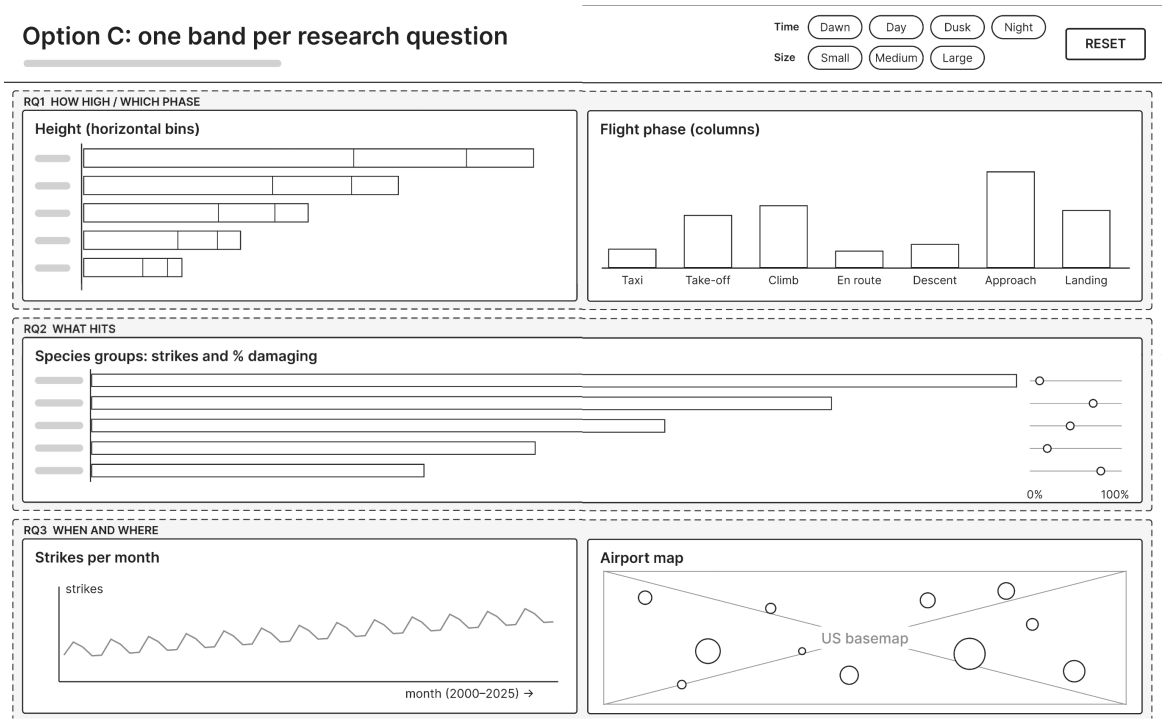


Figure 3: Option C: one band per research question.

Annotations for Option A (Fig. 1).

1. Title states the main insight and the subtitle restates the finding for the current selection and updates with every filter.
2. KPI strip (strikes, % damaging, etc): visibly largest numbers, so they are read first and is recomputed for every filter.
3. Each airport is a point at coordinates: circle area = strikes, fill lightness = %damage. Click, shift-click, or box-select filters all views.
4. Height profile: up = higher. Bar length = strikes, segments = damage level. A vertical brush filters all views.
5. Phase bars stay in flight order (ordinal, not sorted). Click phases to filter and stacked by damage level.
6. Timeline: time runs left to right. The dashed box filters all views; axes stay fixed while filtering.
7. Species: bar = strikes (sorted); dot = % damage on an aligned 0–100% axis. Direct labels. Click to filter.
8. Time-of-day and animal-size chips are global filters; RESET clears every brush and selection.

4 Channel Justification Table

Data variable	Data type	Mark	Channel	Principle	Justification
Strike count (per bin / group / month)	Quantitative	Bar (line in V4)	Length on a common baseline (V2, V3, V5); y-position (V4)	Effectiveness	The main measure in every question, so it gets the strongest channel.
Height AGL (binned)	Ordinal (from Q)	Bar	Vertical position, upward	Convention / mental model	"Up = higher" matches intuitive grasp of altitude. Bins are 1,000 ft wide up to 5,000 ft, and the 0 ft bin is labeled.
Phase of flight	Ordinal	Bar	Vertical position, in flight order	Expressiveness	Kept in the order a flight follows, so the axis matches the data's real order.
Damage level	Ordinal	Bar segment, point fill	Color luminance (sequential Oranges, 4 steps)	Expressiveness	An ordered channel for ordered data. L* falls steadily, so it passes the greyscale test.
% damaging (per group / airport)	Quantitative	Point	x-position on an aligned 0–100% scale (V5); fill luminance (V1)	Effectiveness / separability	Position gives an exact comparison in V5. On the map, x/y are already used for location, so luminance gives a rough reading and tooltips give exact values.
Species group	Categorical	Bar row	Spatial region + direct label	Expressiveness	Spatial grouping is the strongest identity channel. Rows are sorted by count with Unidentified and Other at the end.
Airport location	Spatial	Point	2-D position	Expressiveness	Geography is the natural layout, so a map is justified here.
Strikes per airport	Quantitative	Point	Area + size legend	Effectiveness (secondary)	It is used for which airports are "big" with exact values in tooltips and the KPI strip.
Month (time)	Quantitative	Line / area	x-position, left→right	Convention	Time flows left to right, and the brushable x-axis feeds the time filter.
Time of day	Ordinal	Toggle chip	Spatial order (Dawn→Night)	Expressiveness	Answers the day/night part of RQ2.
Animal size	Ordinal	Toggle chip	Spatial order (Small→Large)	Expressiveness	Ordered chips match the ordered data. Damage rises from small to large, which supports RQ2.

References

FAA Wildlife Strike Database and field list, https://wildlife.faa.gov. FAA/USDA, *Wildlife Strikes to Civil Aircraft in the United States, 1990–2024* (summarized in General Aviation News, Aug. 2025).